

COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - Week 1.2 submission template

Questions

3 Use the logical equivalences in Sections 2.1 and 2.2 of the lecture notes to show that

$$\neg(\exists x \forall y (P(x) \wedge ((Q(x) \vee S(y)) \Rightarrow D(x,y))))$$

is equivalent to

$$\forall x \exists y (\neg P(x) \vee (((Q(x) \vee S(y))) \wedge \neg D(x,y)))$$

At each step, state which equivalence you are using.

Write your solution in this box:

8 The Fitch diagram below shows a proof that

$$P_1 \vee (P_2 \wedge P_3) \therefore (P_1 \vee P_2) \wedge (P_1 \vee P_3)$$

is a valid argument. Fill in all of the justifications on the right hand side.

1. $P_1 \vee (P_2 \wedge P_3)$	_____
2. P_1	_____
3. $P_1 \vee P_2$	_____
4. $P_1 \vee P_3$	_____
5. $(P_1 \vee P_2) \wedge (P_1 \vee P_3)$	_____
6. $P_1 \Rightarrow ((P_1 \vee P_2) \wedge (P_1 \vee P_3))$	_____
7. $P_2 \wedge P_3$	_____
8. P_2	_____
9. P_3	_____
10. $P_1 \vee P_2$	_____
11. $P_1 \vee P_3$	_____
12. $(P_1 \vee P_2) \wedge (P_1 \vee P_3)$	_____
13. $(P_2 \wedge P_3) \Rightarrow ((P_1 \vee P_2) \wedge (P_1 \vee P_3))$	_____
14. $(P_1 \vee P_2) \wedge (P_1 \vee P_3)$	_____